

- 4 (a) gravitational PE is energy of a mass due to its position in a gravitational field B1
 elastic PE energy stored (in an object) due to (a force) changing its shape /
 deformation / being compressed / stretched / strained B1 [2]
- (b) (i) 1. kinetic energy = $\frac{1}{2} mv^2$ C1
 $= \frac{1}{2} \times 0.065 \times 16^2 = 8.3(2) \text{ J}$ A1 [2]
2. $v^2 = 2gh$ OR $PE = mgh$ C1
 $h = 16^2 / (2 \times 9.81) = 13(.05) \text{ m}$ A1 [2]

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- (ii) speed at $t = \frac{1}{2}$ total time = $8 \text{ (ms}^{-1}\text{)}$ or total $t = 1.63$ or $t_{1/2} = 0.815 \text{ s}$ C1
KE is $\frac{1}{4}$ or h at $t_{1/2} = 9.78 \text{ (m)}$ C1
and PE is $\frac{3}{4}$ of max ratio = 3 or ratio = $9.78 / 3.26 = 3$ A1 [3]
- (iii) time is less because (average) acceleration is greater OR average force is greater B1 [1]